

Independent Study

AIEA Lab task 1

Dr. Leilani Gilpin

Objectives

- Exploring project directions post-offboarding auditing

Intro

This has been a difficult week. I have a lot of ideas in Neuro-symbolic AI that I would love to explore. This week I tried exploring pre-existing projects, brainstorming with Leilani, and my own ideas.

Exploring Pre-Existing Projects

I have taken a look at all the projects that I could join. I was mainly interested in LLM Law. That project was interesting because they have done a lot of the work that I have been doing throughout the tasks. I talked to Sebastian and Alex during the working sessions and spent time exploring and understanding their codebase. Their idea is very interesting and for semantic parsing from NL → Prolog they are fine tuning an LLM to do so. The project seems super fun. However I am not the most interested in Law and the dataset is not my personal favorite.

Brainstorming With Leilani

Leilani suggested an interesting project that involves transcribing notes and symbols in notes and comparing other people's notes together. I find this route fairly interesting. I can explore things like VLMs, OCR, NeuroSymbolic translations, reasoning models, research papers, etc. I am interested in getting better at creating Neurosymbolic pipelines.

My Own Ideas

I have been thinking a lot about reasoning models. Simply, they are models that use COT. However, that is not everything. These models also use forms of self verification and such. These models are engineered to be more reasoning based. The issue is this comes at a higher cost. Reasoning models are much more expensive and cost a lot to implement. They also use up a lot of resources. If I can effectively prove Neurosymbolic AI to beat reasoning models on reasoning tasks and are more resource efficient then it could be worth my time.

Conclusion

I have done a lot of exploration and talked to a lot of people and skimmed through a lot of papers this week. I need to explore more and fill gaps in my knowledge. Potentially, starting with a literature review next.